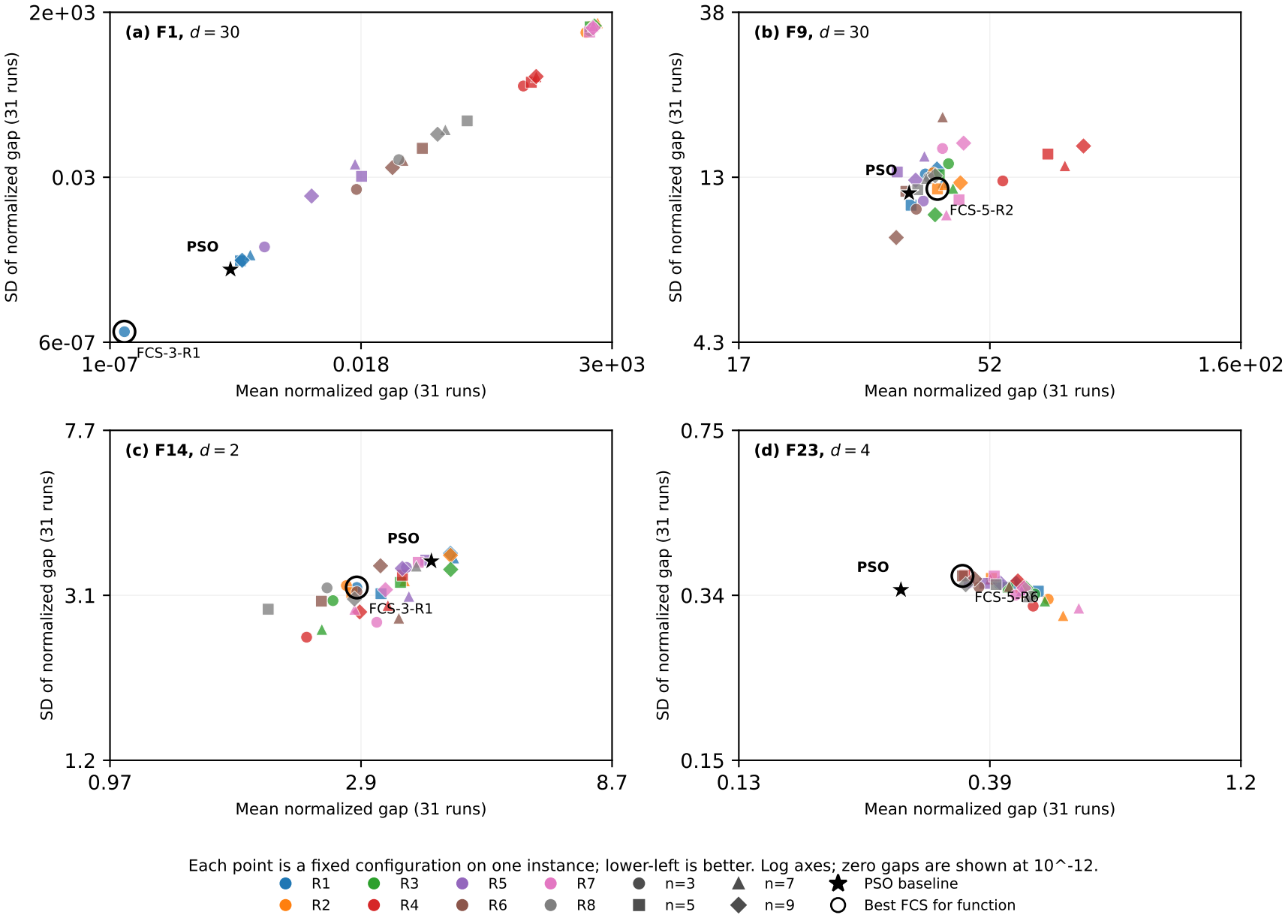
(a) F1, d = 30
PSO
FCS-3-R1
SD of normalized gap (31 runs)
Mean normalized gap (31 runs)
2e+03
0.03
6e-07
1e-07
0.018
3e+03
(b) F9, d = 30
PSO
FCS-5-R2
38
13
4.3
17
52
1.6e+02
(c) F14, d = 2
PSO
FCS-3-R1
7.7
3.1
1.2
0.97
2.9
8.7
(d) F23, d = 4
PSO
FCS-5-R6
0.75
0.34
0.15
0.13
0.39
1.2
Each point is a fixed configuration on one instance; lower-left is better. Log axes; zero gaps are shown at 10^-12.
R1
R2
R3
R4
R5
R6
R7
R8
n=3
n=5
n=7
n=9
PSO baseline
Best FCS for function